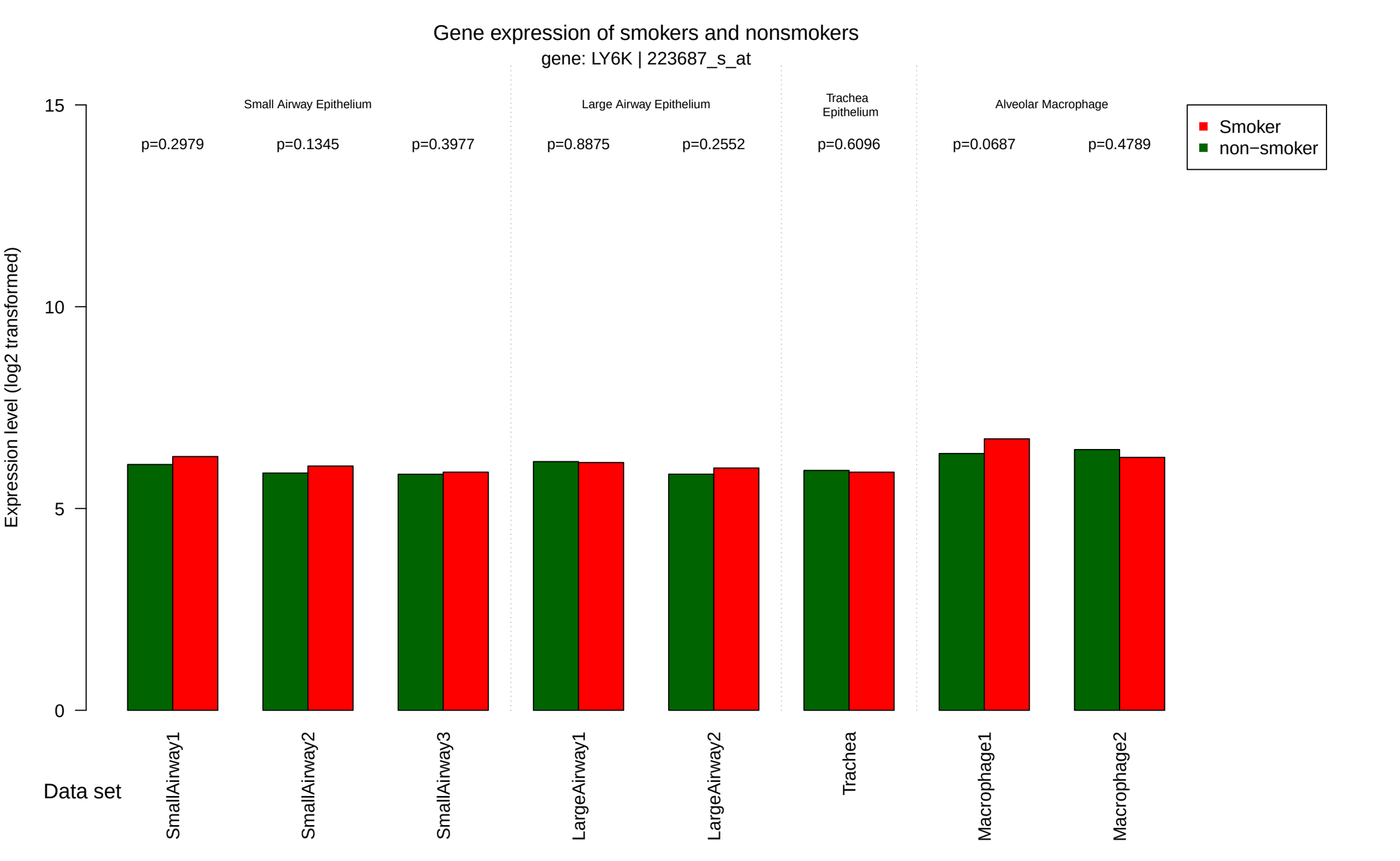
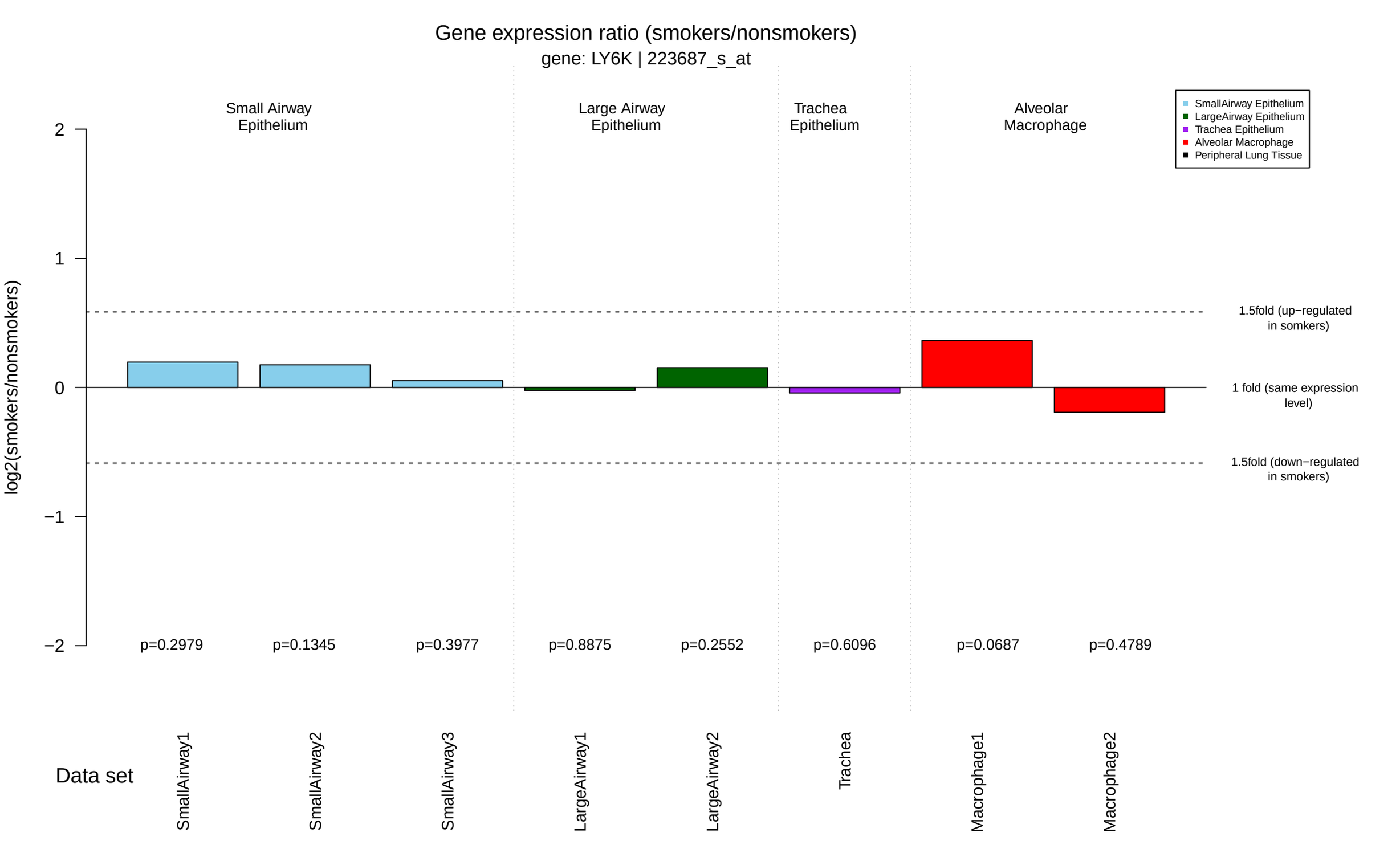
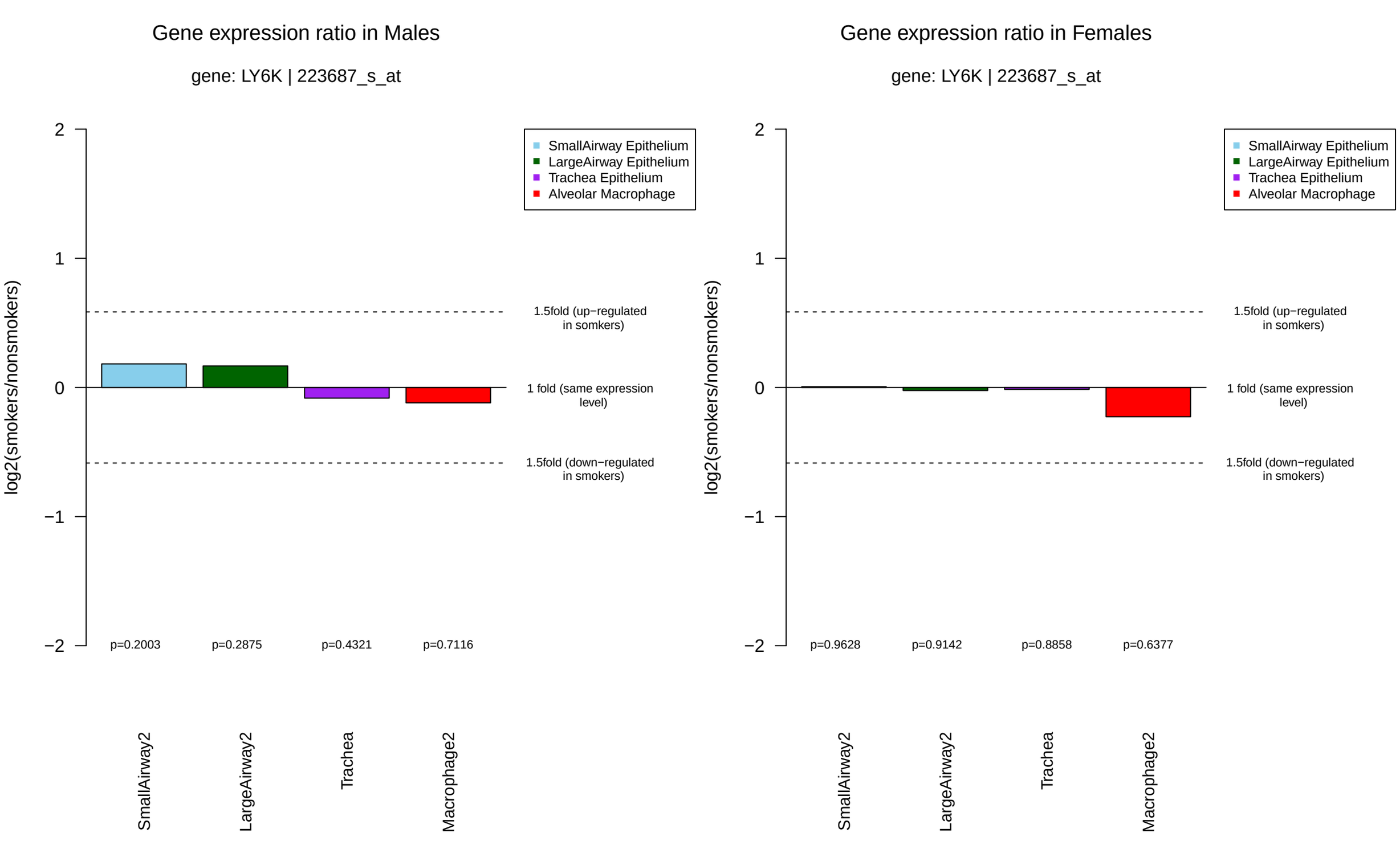
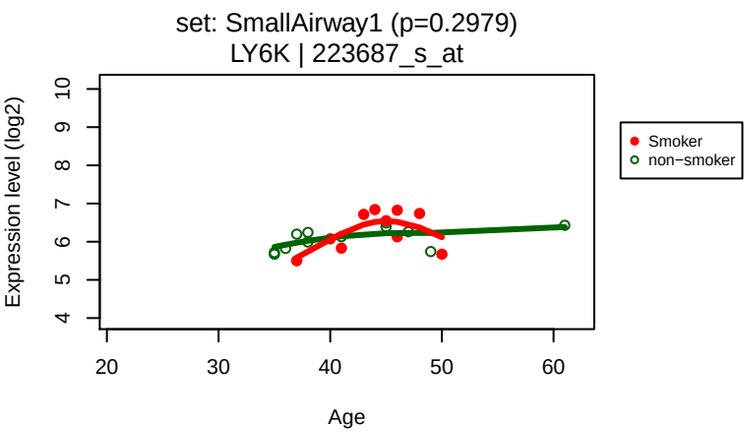
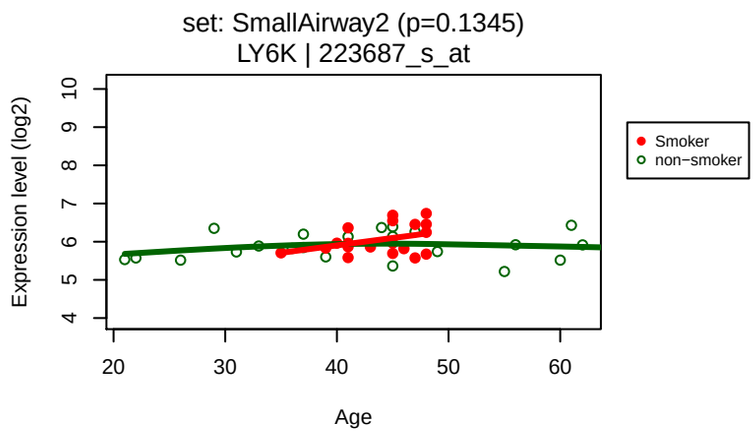
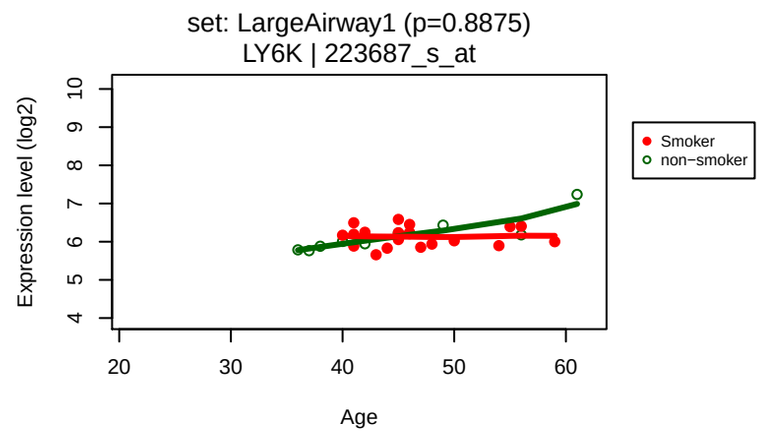

Gene expression of smokers and nonsmokers
gene: LY6K | 223687_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2979
p=0.1345
p=0.3977
p=0.8875
p=0.2552
p=0.6096
p=0.0687
p=0.4789
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

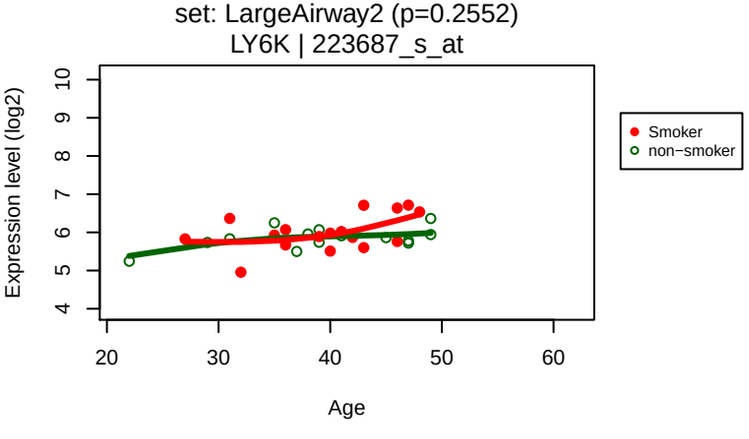

Gene expression ratio (smokers/nonsmokers)
gene: LY6K | 223687_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2979
p=0.1345
p=0.3977
p=0.8875
p=0.2552
p=0.6096
p=0.0687
p=0.4789
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

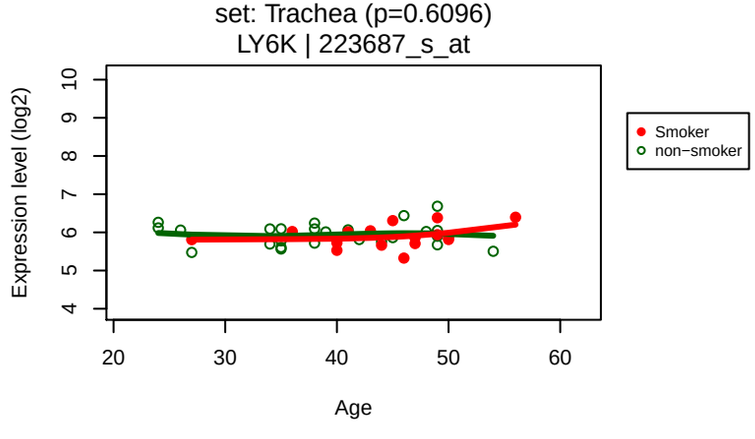

Gene expression ratio in Males
gene: LY6K | 223687_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2003
p=0.2875
p=0.4321
p=0.7116
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LY6K | 223687_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9628
p=0.9142
p=0.8858
p=0.6377
SmallAirway2
LargeAirway2
Trachea
Macrophage2

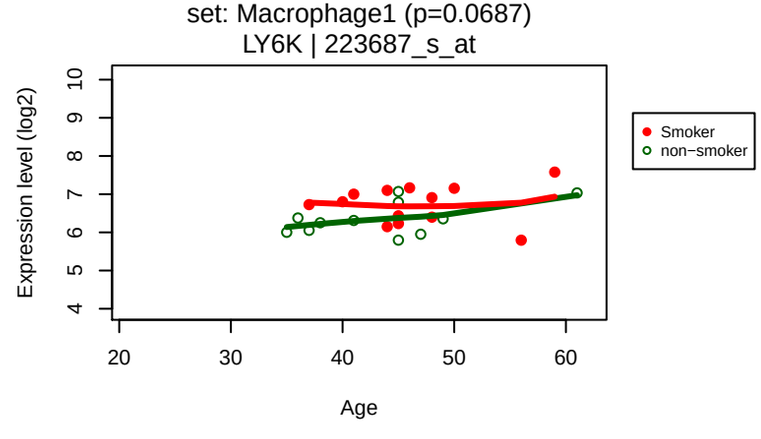

set: SmallAirway1 (p=0.2979)
LY6K | 223687_s_at
Expression level (log2)
Age
Smoker
non−smoker

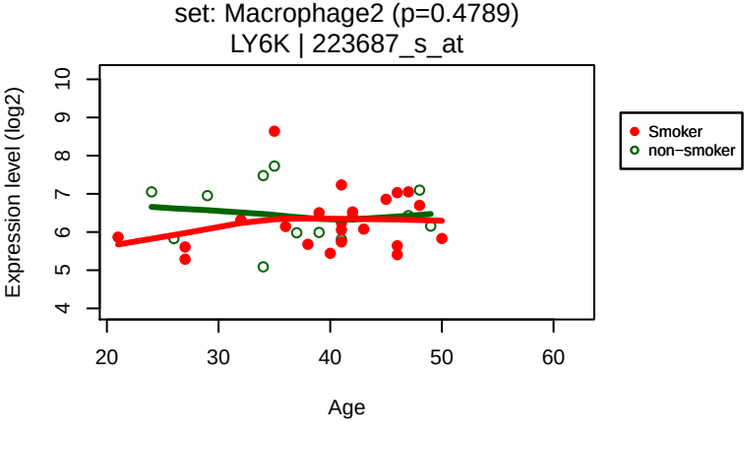

set: SmallAirway2 (p=0.1345)
LY6K | 223687_s_at
Expression level (log2)
Age
Smoker
non−smoker

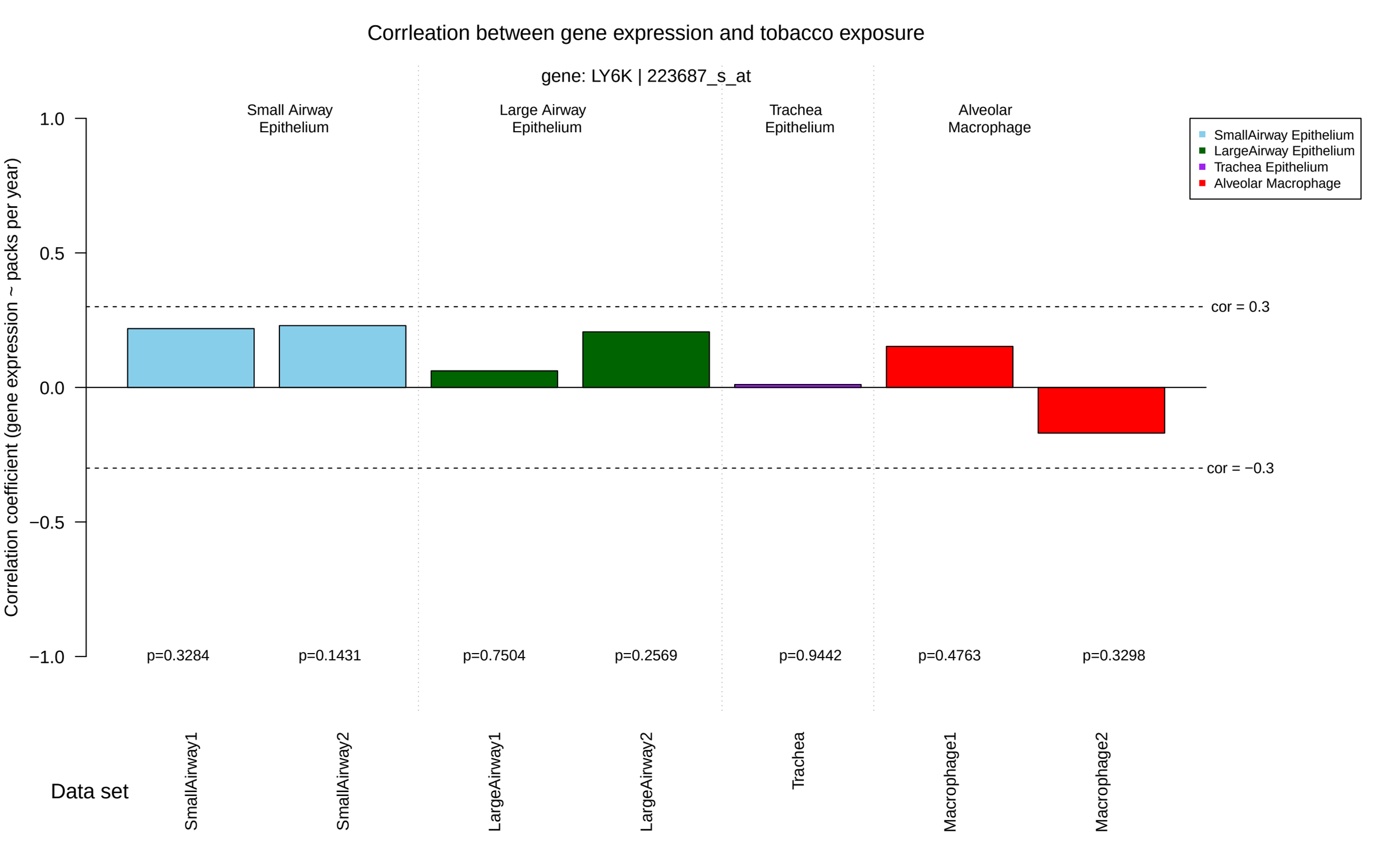

set: LargeAirway1 (p=0.8875)
LY6K | 223687_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2552)
LY6K | 223687_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6096)
LY6K | 223687_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0687)
LY6K | 223687_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4789)
LY6K | 223687_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LY6K | 223687_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3284
p=0.1431
p=0.7504
p=0.2569
p=0.9442
p=0.4763
p=0.3298
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2